5/21 - §8. Differential Entropy

Let X be random variable defined by

$$X = x \quad \text{with probability } p(x)$$

$$H(X) = -\sum_x p(x) \log p(x)$$

Let X be random variable

$\boxed{\text{Def}}$ Cumulative distribution fcn $F(x) := Pr(X \leq x)$

$\boxed{\text{Def}}$ X is continuous $\Leftrightarrow$ F is continuous

$\boxed{\text{Def}}$ If $F'(x)$ exists, then $f(x) = F'(x)$ is called "probability density fcn"

$\boxed{\text{Def}}$ $h(X) := -\int_S f(x) \log f(x)\, dx$

$$S := \{x \mid f(x) > 0\} \quad ; \text{ support set of } \begin{cases} f. \\ X. \end{cases}$$

$\boxed{\text{Example}}$ $X \sim N(0, \sigma)$, $f(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}}$

$$h(X) = -\int_{-\infty}^{\infty} f(x) \log f(x)\, dx$$

$$= -\int_{-\infty}^{\infty} f(x) \left(\frac{-x^2}{2\sigma^2} \log e - \frac{1}{2} \log(2\pi\sigma^2) \right) dx$$

$$= \frac{\log e}{2\sigma^2} \operatorname{var}(X) + \frac{1}{2} \log(2\pi\sigma^2)$$

$$= \frac{1}{2} \log e + \frac{1}{2} \log(2\pi\sigma^2) = \frac{1}{2} \log(2\pi e\sigma^2)$$

In Ch.3. AEP (By 윤재영).

$X_1, \cdots, X_n$; i.i.d $\sim p(x)$, $X \sim p(x)$

$\boxed{\text{Thm 3.1.1}}$ $-\frac{1}{n}\log p(X_1, \cdots, X_n) \xrightarrow[\text{in probability}]{} H(X)$

$$A_\varepsilon^{(n)} := \left\{ (x_1, \cdots, x_n) \,\middle|\, 2^{-n(H+\varepsilon)} \le p(x_1, \cdots, x_n) \le 2^{-n(H-\varepsilon)} \right\}$$

$\boxed{\text{Thm 8.2.1}}$ Let $X_1, \cdots, X_n$ be i.i.d $\sim f(x)$

$$-\frac{1}{n}\log f(X_1, \cdots, X_n) \longrightarrow h(X) \quad \text{in probability}$$

$$\left(f(x_1, \cdots, x_n) = \prod_{i=1}^{n} f(x_i) \right)$$

$\boxed{\text{Def}}$ $A_\varepsilon^{(n)} := \left\{ (x_1, \cdots, x_n) : \left| -\frac{1}{n}\log f(x_1, \cdots, x_n) - h(X) \right| \le \varepsilon \right\}$

$\boxed{\text{Thm 8.2.2}}$ (Conti version of Thm 3.1.2)

1. $\Pr(A_\varepsilon^{(n)}) > 1 - \varepsilon$ for $n \gg 1$
2. $\mathrm{Vol}(A_\varepsilon^{(n)}) \le 2^{n(h+\varepsilon)}$ for all n

   $\mathrm{Vol}(A) = \int_A dx_1 \cdots dx_n$
3. $\mathrm{Vol}(A_\varepsilon^{(n)}) \ge (1-\varepsilon) \cdot 2^{n(h-\varepsilon)}$ for $n \gg 1$

pf)

2.
$$\begin{aligned} 1 &= \int_{S^n} f(x_1, \cdots, x_n)\, dx_1 \cdots dx_n \\ &\ge \int_{A_\varepsilon^{(n)}} 2^{-n(h+\varepsilon)}\, dx_1 \cdots dx_n \\ &= 2^{-n(h+\varepsilon)}\, \mathrm{Vol}(A_\varepsilon^{(n)}) \end{aligned}$$

3. $$1-\varepsilon \le \int_{A_\varepsilon^{(n)}} f(x_1,\cdots,x_n)\,dx_1\cdots dx_n$$
$$\le \int_{A_\varepsilon^{(n)}} 2^{-n(h-\varepsilon)}\,dx_1\cdots dx_n$$
$$= 2^{-n(h-\varepsilon)}\,\mathrm{Vol}(A_\varepsilon^{(n)}) \qquad (n \gg 1)$$

$\boxed{\text{Thm 8.2.3}}$ (Exercise 3-11)

$A_\varepsilon^{(n)}$ is the smallest volume set with probability $\ge 1-\varepsilon$ to first order in the exponent.

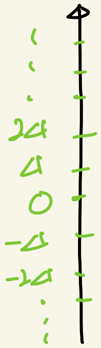
pf) Let $B_\varepsilon^{(n)}$ be the smallest volume set with probability $\ge 1-\varepsilon$

$$1-2\varepsilon \le \Pr(A_\varepsilon^{(n)} \cap B_\varepsilon^{(n)})$$
$$= \int_{A_\varepsilon^{(n)} \cap B_\varepsilon^{(n)}} f(x_1,\cdots,x_n)\,dx_1\cdots dx_n$$
$$\le \int_{A_\varepsilon^{(n)} \cap B_\varepsilon^{(n)}} 2^{-n(h-\varepsilon)}\,dx_1\cdots dx_n$$
$$\le 2^{-n(h-\varepsilon)}\,\mathrm{Vol}(B_\varepsilon^{(n)})$$
$$(1-2\varepsilon)\,2^{n(h-\varepsilon)} \le \mathrm{Vol}(B_\varepsilon^{(n)})$$
$$(1-\varepsilon)\,2^{n(h-\varepsilon)} \le \mathrm{Vol}(A_\varepsilon^{(n)}) \le 2^{n(h+\varepsilon)}$$
$$0 \le \frac{1}{n}\log\frac{\mathrm{Vol}(A_\varepsilon^{(n)})}{\mathrm{Vol}(B_\varepsilon^{(n)})} \le \frac{1}{n}\log\frac{2^{n(h+\varepsilon)}}{(1-2\varepsilon)\,2^{n(h-\varepsilon)}} \xrightarrow[n\to\infty]{} 0$$

$\square$

Digital data is quantized and has minimal unit Δ

X ; conti r.v.



$$\Pr(i\Delta \le X < (i+1)\Delta)$$
$$= \int_{i\Delta}^{(i+1)\Delta} f(x)\,dx = f(x_i)\cdot\Delta =: p_i$$

for some $i\Delta \le x_i \le (i+1)\Delta$

$\boxed{\text{Def}}$ X^Δ ; quantized X

$X^\Delta = x_i$ with probability p_i

$$H(X^\Delta) = -\sum p_i \log p_i = -\sum f(x_i)\Delta \log f(x_i) - \sum f(x_i)\Delta \log \Delta$$
$$= -\sum f(x_i)\Delta \log f(x_i) - \log \Delta$$

$$H(X^\Delta) + \log \Delta = -\sum f(x_i)\Delta \log f(x_i)$$
$$\Big\downarrow \Delta \to 0$$
$$-\int f(x) \log P(x)\,dx = h(X)$$

$\boxed{\text{Thm 8.3.1}}$ $H(X^\Delta) + \log\Delta \longrightarrow h(X)$ as $\Delta \to 0$

$\boxed{\text{Example 8.3.1}}$ $X \sim \text{Unif}[0,1]$, $\Delta = 2^{-n}$

$\Rightarrow$ $h(X) = 0$, $H(X^\Delta) = n \xrightarrow[n\to\infty]{} \infty$

In Ch 2. many concepts. like relative entropy, mutual information...

$\boxed{\text{Def}}$ $h(x_1, \cdots, x_n) = -\int f(x_1, \cdots, x_n) \log f(x_1, \cdots, x_n)\, dx_1 \cdots dx_n$

$\boxed{\text{Def}}$ (Conditional entropy)

$$H(X|Y) = \sum_x p(x) H(Y|X=x)$$
$$= -\sum_x p(x) \sum_y p(y|x) \log p(y|x)$$
$$= -\sum_{x,y} p(x,y) \log p(y|x)$$

$\boxed{\text{Def}}$ $h(X|Y) = -\int f(x,y) \log f(x|y)\, dx dy$

$$= -\int f(x,y) \left(\log f(x,y) - \log f(y) \right) dx dy$$
$$= h(X,Y) - h(Y)$$

$\boxed{\text{Def}}$ $D(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$ (relative entropy)

$\boxed{\text{Def}}$ $D(f\|g) = \int_S f \log \frac{f}{g}$

$S :=$ (support set of f) $\subseteq$ (support set of g)

$\boxed{\text{Def}}$ (mutual information) $I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$

$\boxed{\text{Def}}$ $I(X;Y) = \int f(x,y) \log \frac{f(x,y)}{f(x)f(y)}\, dx dy$

$\boxed{\text{Thm 8.6.1}}$ $D(f\|g) \geq 0$,

equality holds iff $f = g$ a.e.

pf) $-D(f\|g) = \int_S f \log \frac{g}{f}$

$$\leq \log \int_S f \cdot \frac{g}{f} \leq \log \int_S g \leq 0$$

$$0 = D(f\|g) = \int_S f \log \frac{f}{g} \iff \frac{f}{g} = 1 \quad \text{a.e.}$$

$\square$

$\boxed{\text{Cor}}$

$$I(X;Y) = \int f(x,y) \log \frac{f(x,y)}{f(x)f(y)}\, dxdy$$

$$= D(f(x,y) \,\|\, f(x)f(y)) \geq 0$$

equality holds $\iff f(x,y) = f(x)f(y)$ a.e

$\iff$ X and Y are independent.

$\boxed{\text{Cor}}$

$$I(X;Y) = \int f(x,y) \log \frac{f(x,y)}{f(x)f(y)}\, dxdy$$

$$= \int f(x,y) \log f(x|y)\, dxdy - \int f(x,y) \log f(x)\, dxdy$$

$$= -h(X|Y) + h(X) \geq 0$$

$$h(X|Y) \leq h(X)$$

$$h(X_1,\cdots,X_n) = -\int f(x_1,\cdots,x_n)\log f(x_1,\cdots,x_n)\,dx_1\cdots dx_n$$

$$f(x_1,\cdots,x_n) = f(x_n \mid x_1,\cdots,x_{n-1})\,f(x_1,\cdots,x_{n-1})$$
$$= f(x_n \mid x_1,\cdots,x_{n-1})\,f(x_{n-1} \mid x_1,\cdots,x_{n-2})\,f(x_1,\cdots,x_{n-2})$$
$$= \cdots = f(x_n \mid x_1\cdots x_{n-1})\,f(x_{n-1} \mid x_1,\cdots,x_{n-2})\cdots f(x_1)$$

$$h(X_1,\cdots,X_n) = \sum_{i=1}^{n} h(X_i \mid X_1,\cdots,X_{i-1}) \le \sum_{i=1}^{n} h(X_i)$$

$\boxed{\text{Example}}$ Let X be random vector $X \sim N(0, K)$

$X = (X_1,\cdots,X_n)$

covariance matrix

$$h(X_1,\cdots,X_n) \le \sum_{i=1}^{n} h(X_i) = \sum_{i=1}^{n} \frac{1}{2}\log(2\pi e K_{ii})$$

$\| \boxed{\text{Thm 8.4.1}}$

$$\frac{1}{2}\log(2\pi e)^n |K|$$

$$|K| \le \prod_{i=1}^{n} K_{ii}$$

$$\boxed{\text{For all positive definite matrix } K \text{, we have } |K| \le \prod_{i=1}^{n} K_{ii}}$$

Hadamard's inequality

$\boxed{\text{Thm 8.6.5}}$ Let X be r.v. with $\begin{cases} \text{zero mean} \\ \text{covariance matrix } K \end{cases}$

$$h(X) \le \frac{1}{2}\log(2\pi e)^n |K|$$

equality holds $\iff X \sim N(0, K)$

pf) Let $\begin{cases} g \text{ be prob. density fcn of } X \\ \phi_K \text{ ——— // ——— of } N(0,K) \end{cases}$

$$0 \le D(g \| \phi_K) = \int g \log \frac{g}{\phi_K} = -h(X) - \int g \log \phi_K$$

$$\phi_K(x) = \frac{1}{(2\pi)^{\frac{n}{2}} |K|^{\frac{1}{2}}} e^{-\frac{1}{2} x^T K^{-1} x}$$

$\log \phi_K$ is quadratic

$$\int g(x) x_i x_j = \mathrm{cov}(X_i, X_j) = K_{ij} = \int \phi_K(x) x_i x_j$$

$$\Rightarrow \int g \log \phi_K = \int \phi_K \log \phi_K$$

$$0 \le -h(X) - \int \phi_K \log \phi_K = -h(X) + \frac{1}{2}\log(2\pi e)^n |K|$$

$$h(X) \le \frac{1}{2} \log (2\pi e)^n |K|$$

equality holds $\iff g = \phi_K$ a.e

$\square$

$\boxed{\text{Cor}}$ Let X be random variable with

$$EX = 0, \quad Var(X) = \sigma$$

$$h(X) \leq \frac{1}{2}\log(2\pi e \sigma) = \frac{1}{2}\log(2\pi e \, Var(X))$$

$$\frac{1}{2\pi e} e^{2h(X)} \leq Var(X)$$